Preparing Banks for Quantum Computing Risks

Using Open Source Strategies for Post-Quantum Readiness

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TD Bank Open Source Program Office

Brief Introduction.

Mark Paulsen.



A Developer in a Previous Life

Enterprise Advocacy Where Open Source Communities Live

Head, Open Source Program Office

Audit, DevOps, Scaled our Open Source Microservices Platform

Balanced Governance and Innovation (AI Especially)



Quantum Credentials.

Collaboration with the Institute for Quantum Computing

Crypto Agility Proof of Concept

Co-Ran Quantum Program and Info Sessions

Contributed to FS-ISAC and Europol report on post-quantum migration

Winner of a Joint Hackathon Submission for Quantum-Safety

Quantum Credentials.



A Neutron Walks Into A Bar And Asks How Much For A Drink. The Bartender Replies, 'For You, No Charge.

Brief Introduction.

Making Banking More Human

6th Largest Bank in North America

28 Million Global Customers

3,300 Communities Received Support

$169 Million Contributed to Communities



Diving Right In.

In this Session, we'll review 4 Questions

That should help Financial Institutions either start, or speed-up the journey of quantum safety

Four Questions

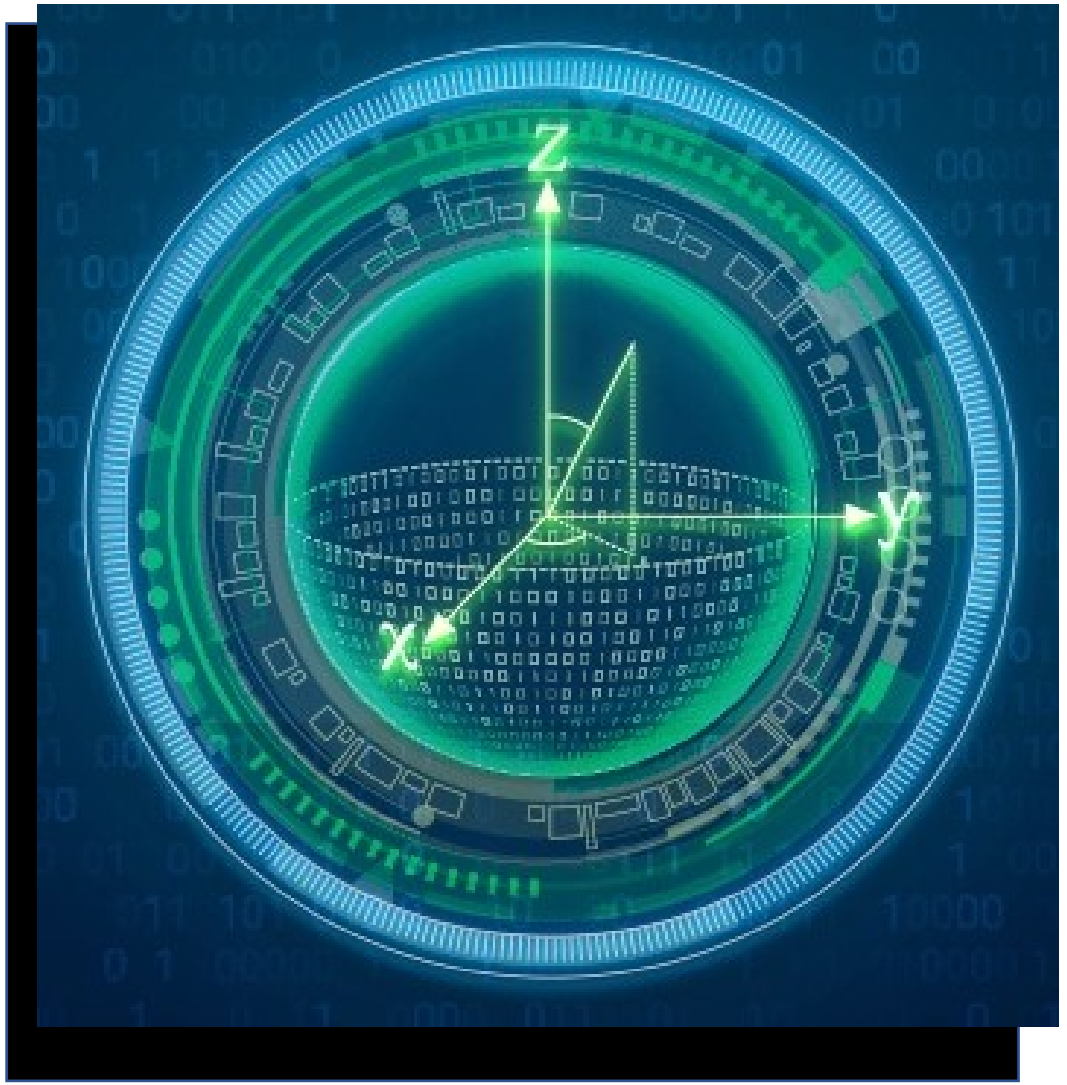
1. What is Quantum Computing?
2. What are the Risks?
3. What does Open Source have to do with Quantum?
4. How can you get started?

1 What is Quantum Computing?

Definition A device that performs Quantum Information processing.

An Introduction to Quantum Computing

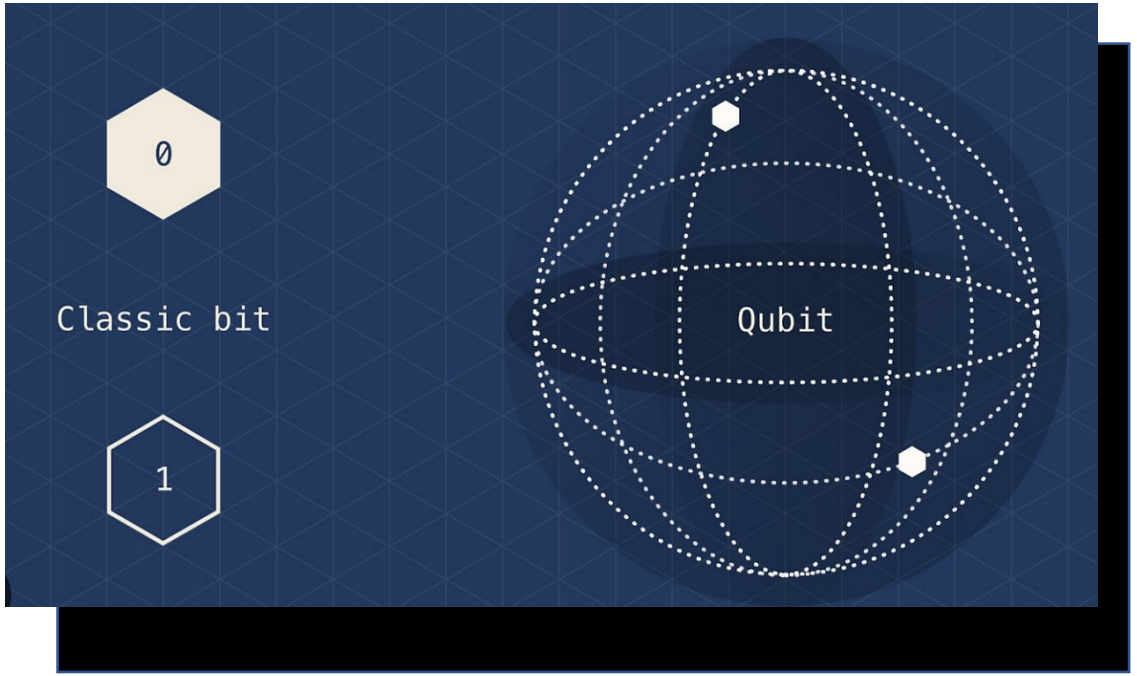
Michele Mosca, Phillip Kaye, and Raymond Laflamme



What is Quantum Information?

Definition

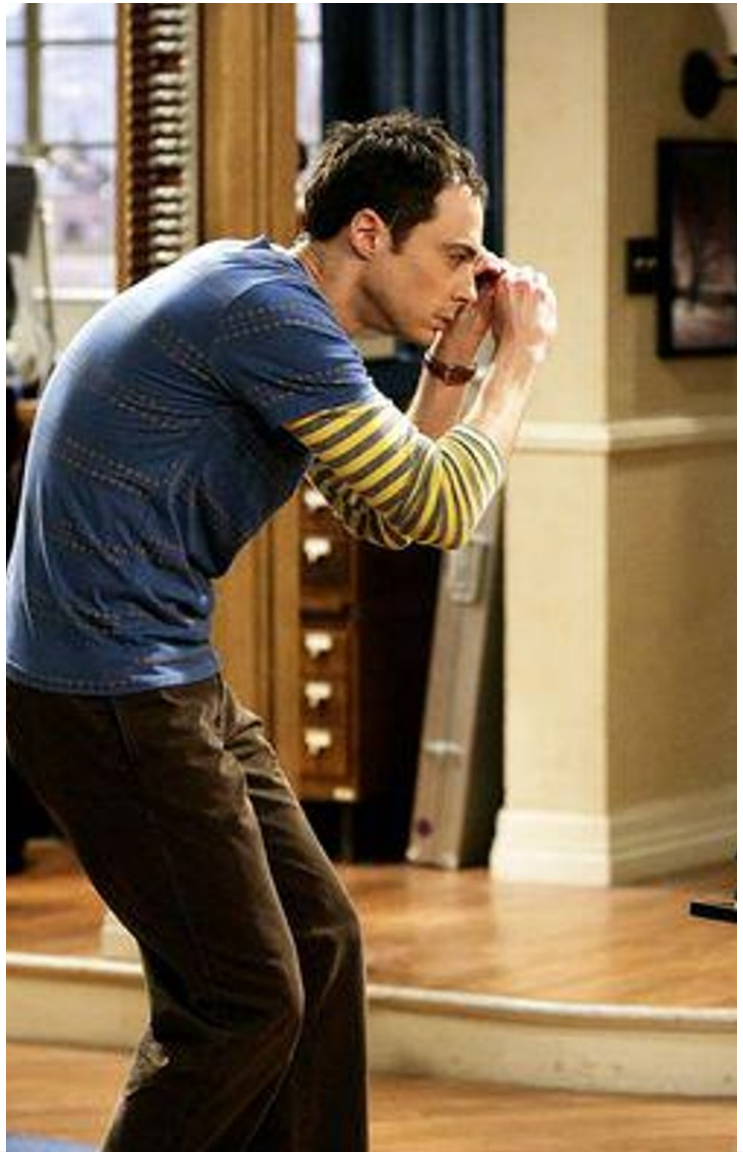
Information carried by a Qubit, which unlike a Classical Bit (0 or 1), can exists in a superposition (0 and 1) at the same time.





Superposition: Combination of all possible states (alive cat / dead cat) at once. Until its measured then it's resolved into a single state (according to a probability).

Power of a Qubit.



Classic Switch with 2 Bits

Can be in **only** one of four different configurations at once

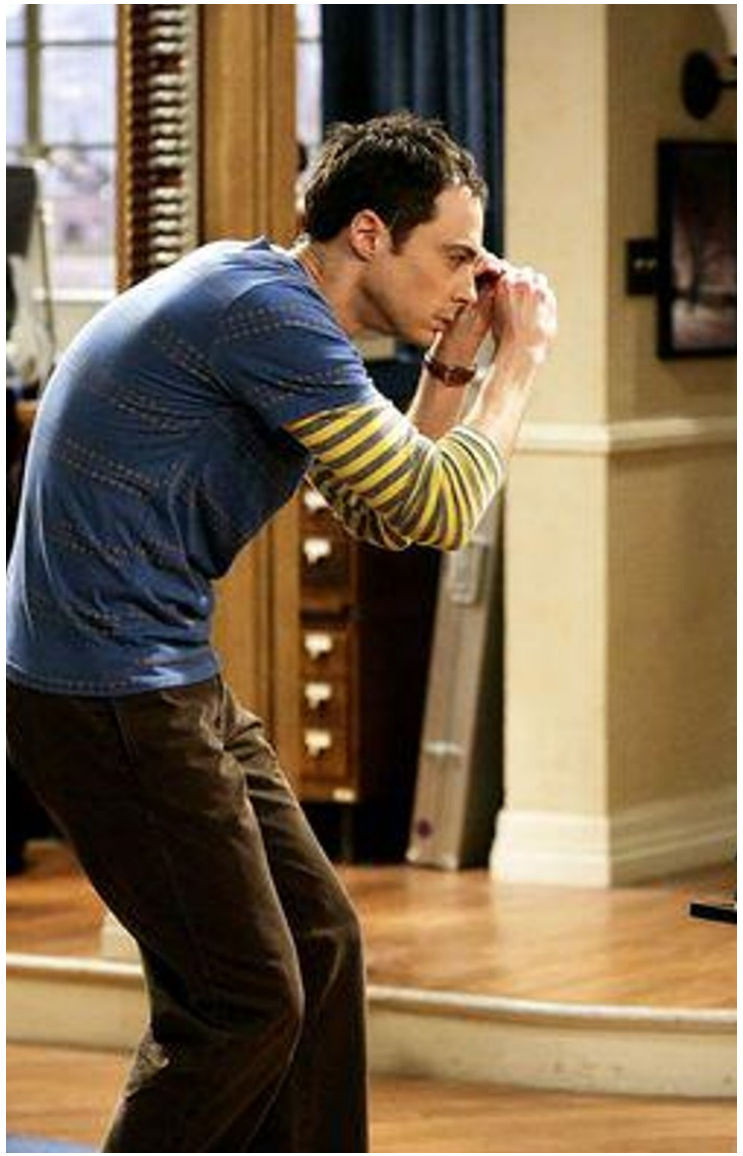
Both Off
0 0

Both On
1 1

Opposite States
0 1

Opposite States
1 0

Power of a Qubit.



Quantum Switch with 2 Qubits

Can be in **all** four different configurations at once

0 0 1 1 0 1 1 0

Quantum Switch with 3 Qubits

Can be in **all eight** different configurations at once

000 001 010 011 100 101 110 111

Quantum Switch with 64 Qubits

Can be ……..in a lot of configurations at once.

2 What are the Risks?



Harvest Now Decrypt Later (HNDL) – stolen data becomes readable in the future





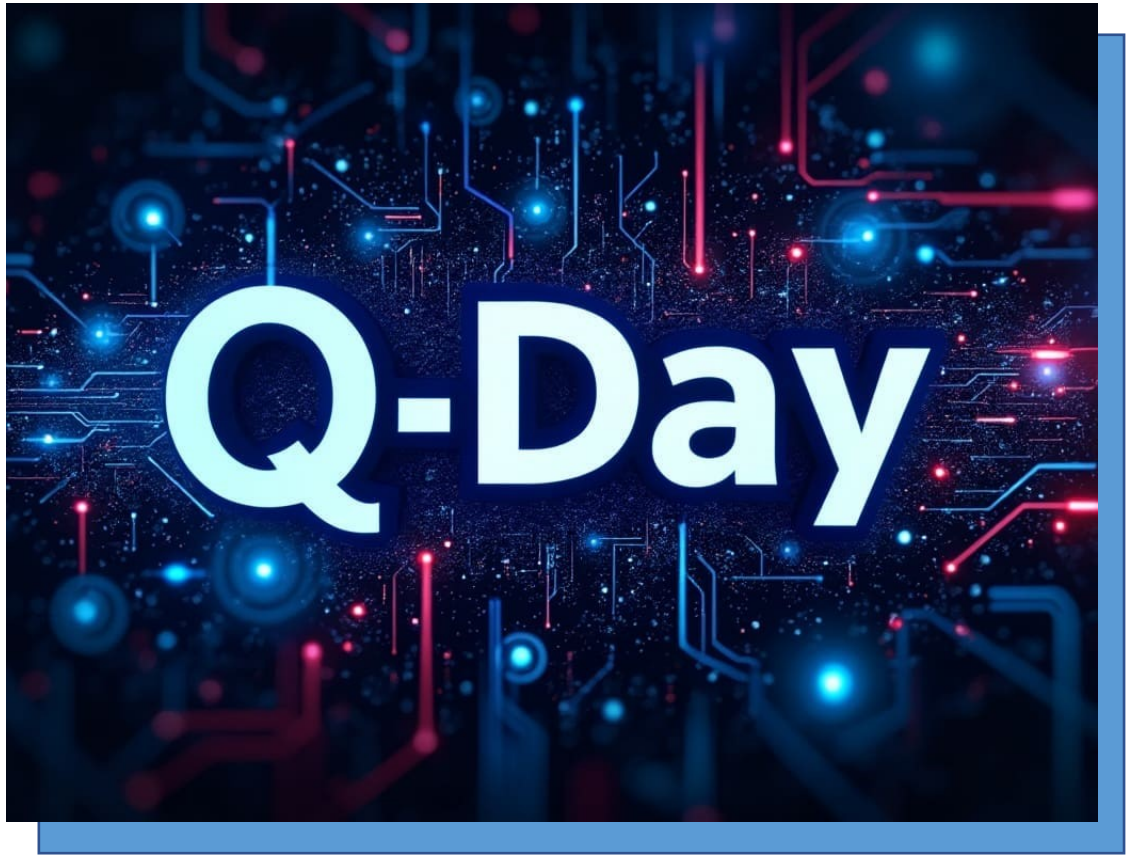
Cryptographic Vulnerability – quantum computers will break current encryption

When is Q-Day?

Definition

The point when quantum computers become powerful enough to break todays widely used public-key cryptography.

Palo Alto Networks



Current Estimate: The 2025 Global Risk Institute Report, with input from Dr. Mosca, says a cryptographically relevant quantum computer is quite possible (28-49%) within the next 10 years.

Mosca's Theorem

Named after the renowned Canadian expert in Quantum Computing and Cryptography which provides a formula for the time required to adopt quantum-resistant algorithms to ensure the long-term confidentiality and integrity of digital data. Provides guidance for managing the risk of "Harvest Now, Decrypt Later" risks.

Theorem

X + Y > Z **If X plus Y is greater than Z you need to worry**

X
The amount of time you need your encrypted data to be secure.

Y
How Long it will take to implement a quantum secure solution into your current infrastructure

Z
How long it will take to develop a sufficiently strong enough scale quantum computer that can break current cryptography.

There are varying estimates, but 2040 would be a conservative timeframe for Z.

Customer Records Data

- **X = 10 Years** – According to the Record Retention Schedule, customer records (retention code DMK-04) have a retention period of 10 years.
- **Y = 5 Years** – Aggressive estimate for how long it will take to implement a quantum secure solution into current infrastructure.
- **Z = 2040** – While the CEO of Nvidia says scalable quantum computing that can break current encryptions is decades away, the Microsoft CEO says its years ago. If we also factor in Google and IBM's estimations, 2040 is a safe and conservative estimate for when a Quantum Computer will be available that can break our current encryptions.

X + Y = 2040 Which is equal to Z
If a Bank started preparing in 2025, the risks should be mitigated

Life Insurance Data

- **X = 30 Years** – According to the Record Retention Schedule, Life Insurance records (retention code FAM-16) have a retention period of 30 years.
- **Y = 5 Years** – Aggressive estimate for how long it will take to implement a quantum secure solution into current infrastructure.
- **Z = 2040** – The year 2040 is a safe and conservative estimate for when a Quantum Computer will be available that can break our current encryptions.

X + Y = 2060 Which is much greater than Z
Action should be taken immediately to mitigate the risk if this data needs to be kept secure long-term.

Shor's Algorithm.

The Near Impossibility of Factoring Larger Numbers – Made Possible (eventually)



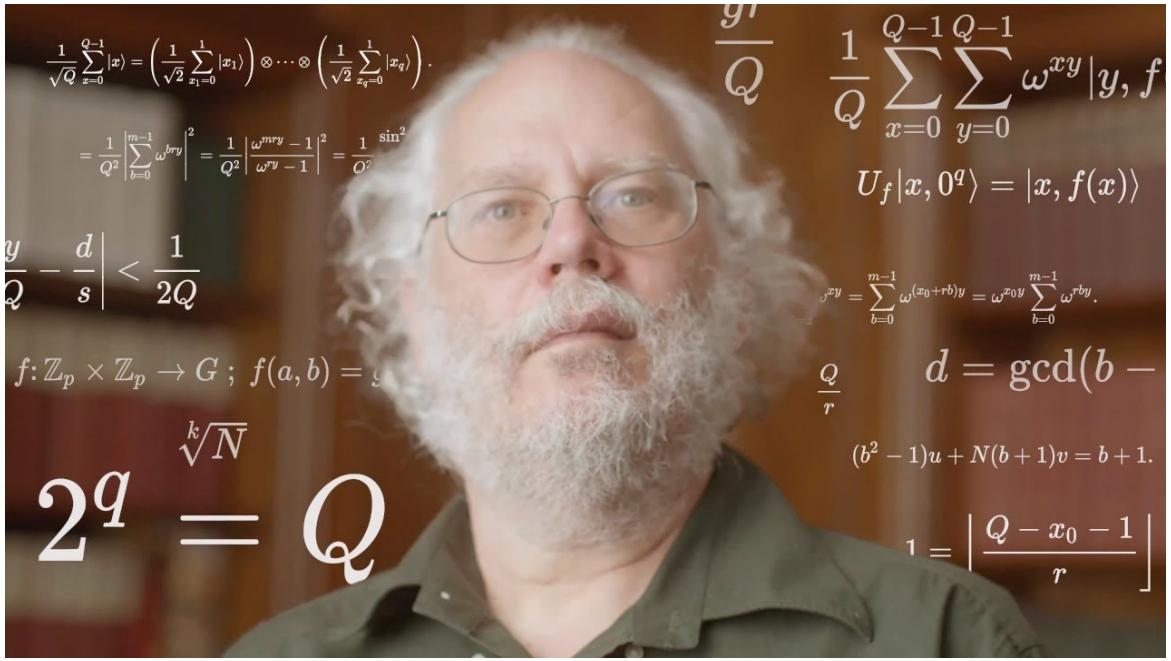
The RSA cipher is used to protect internet traffic and the most important military, diplomatic, and commercial communications. If someone wanted to challenge strong RSA encryption, they would need a way to efficiently factor extremely large numbers. Because it is theoretically impossible to factor very large numbers efficiently, even for modern super computers, information using RSA is deemed safe – for now.

In 1994, Peter Shor created an Algorithm (aka function/program) that leverages a repeatable mathematical wave pattern to efficiently factor numbers.

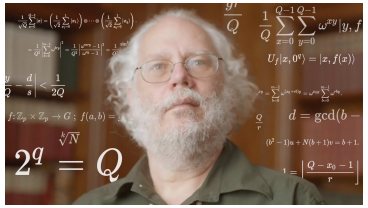
With this algorithm, you would take the public key from the RSA cipher, which is available to everyone, feed it into the Algorithm, and you would be able to find the RSA private key – which means you'd be able to decrypt anything the cipher has encrypted.

Thankfully, to run Shor's Algorithm on an RSA cipher you would need a Quantum Computer running nearly 1 million Qubits. Currently, Quantum Computers have Qubits ranging in the dozens or hundreds.

We have time to prepare – but not that much time!



A bit more on Shor's.



Shor's Algorithm: Spot the Repeat

It races through a list of numbers, watching for the spot where the pattern starts repeating.

THE NUMBER SEQUENCE

1 2 4 8 1 2 4 8 1 2 4 8 ...

repeats every 4 steps

Find the repeat → calculate the two factors instantly.

Example shown: cracking the number 15, which unlocks 3 × 5

A full-scale Quantum Computer that can run Shor's will be great at finding patterns. This opens up a ton of opportunities.

3 What does Open Source have to do with Quantum?

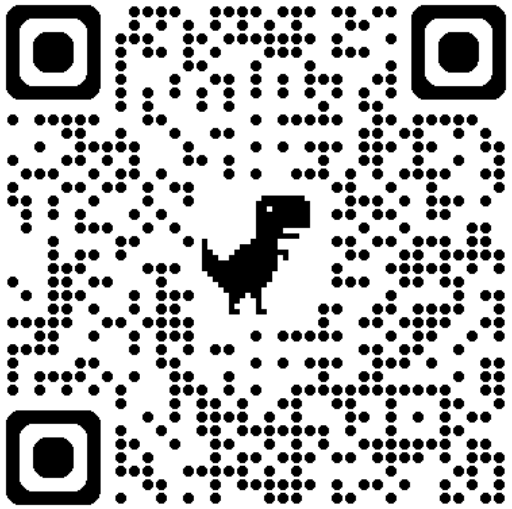
Sometimes explaining Open Source Risks, and Quantum Risks, makes us look like Don Quixote Attacking Windmills

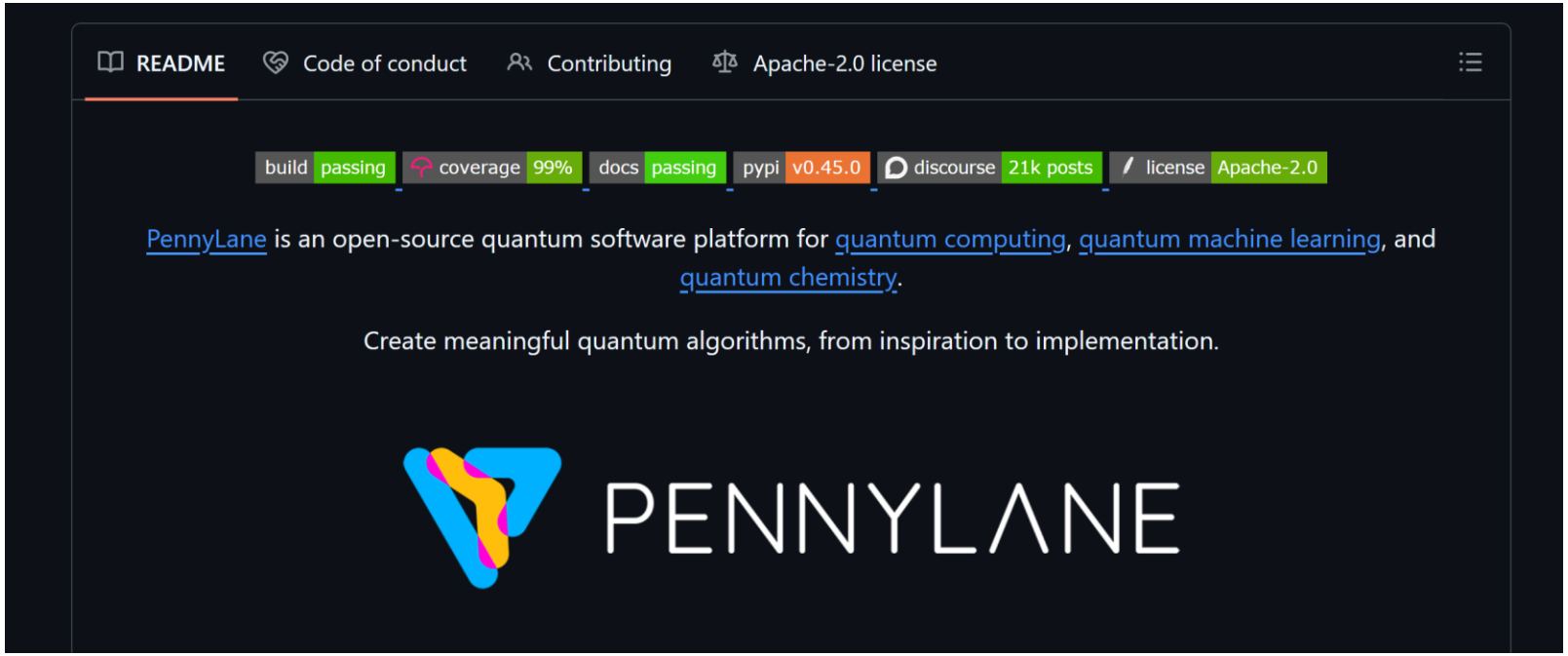


Have you ever heard – "We don't use Open Source" or "We have decades to prepare for Quantum Compute."

Open Source Quantum Computing Development.

Software library for programming quantum computers





Open Source Quantum-resistant Cryptographic Algorithm Project.

Open Source project that aims to support the transition to quantum-resistant cryptography.

OQS is part of the Linux Foundation's Post-Quantum Cryptography Alliance.



New Canadian Forum for Digital Infrastructure Resilience (CFDIR) Working Group .

Collaborative approach to help our audit and risk friends prepare for Quantum Risks.

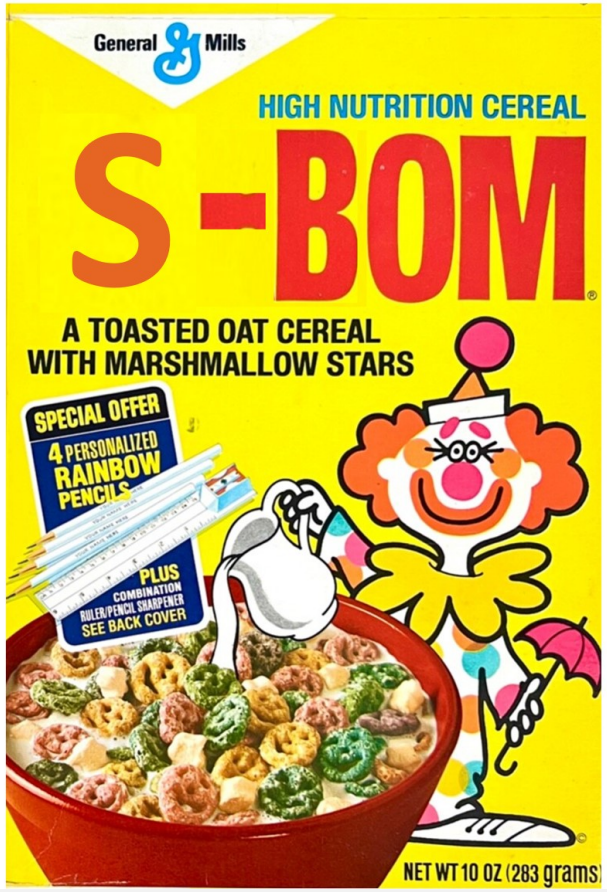


Open Quantum Design.

Full-stack, open-source quantum computing systems



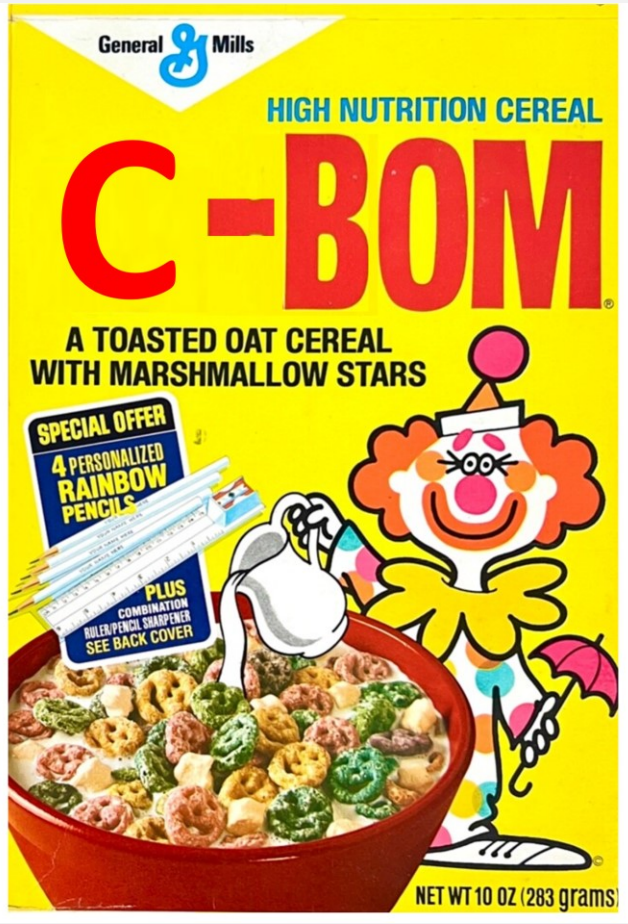
Inventories and Bill of Materials are Key – you can't govern or what you don't know.



Software Bill of Materials

Nutrition Facts
Serving Size 30 Model Deployments

Amount Per Deployment		
Regulatory Alignment	90 %	90 %
Controls from Audit	10	10
	% Coverage	
Immutability	90 %	90 %
Saturated Fat 0g	0%	0%
Trans Fat 0g	0%	0%
Open Source Licenses	20%	20%
Application Mapping	2%	8%
Model Specifications	8%	10%
Dietary Fiber 5g	20%	20%
Sugars 5g		
Dependencies		
Vitamin A	0%	4%
Vitamin C	10%	15%
Calcium	0%	15%
Iron	2%	2%



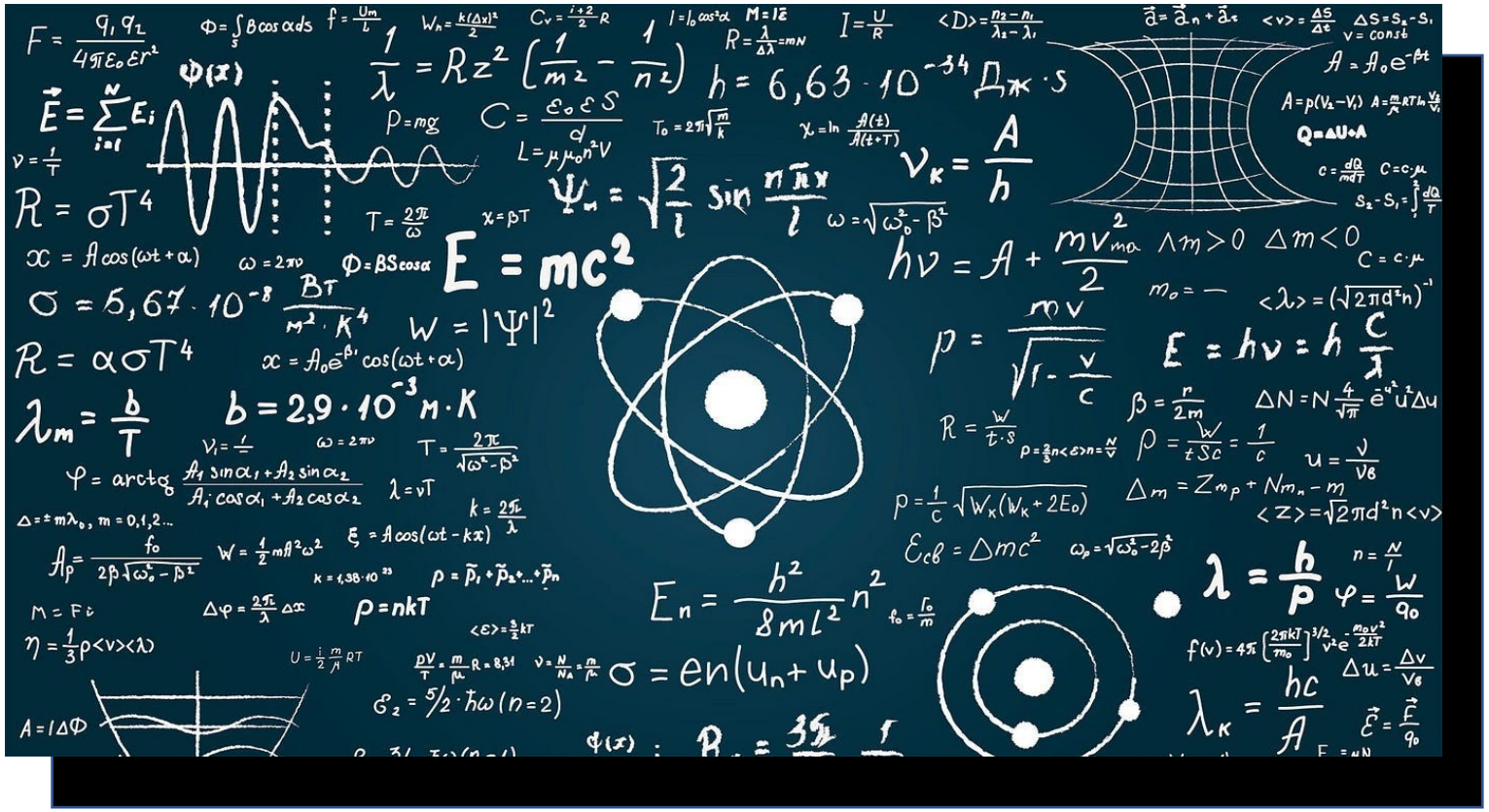
Crypto Bill of Materials

Nutrition Facts
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Regulatory Alignment	90 %	90 %
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Saturated Fat 0g	0%	0%
Trans Fat 0g	0%	0%
Open Source Licenses	20%	20%
Application Mapping	2%	8%
Model Specifications	8%	10%
Dietary Fiber 5g	20%	20%
Sugars 5g		
Dependencies		
Vitamin A	0%	4%
Vitamin C	10%	15%
Calcium	0%	15%
Iron	2%	2%

4 How can you get started?

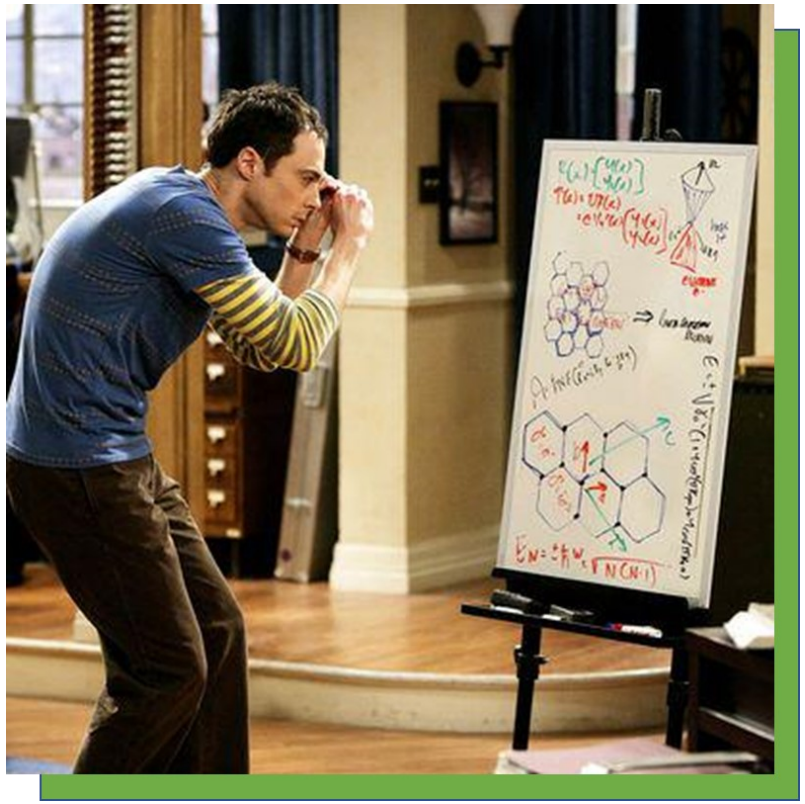
Do you need to go back to school and get a PhD in Quantum Mechanics?





Thankfully, no. The problems Financial Institutions need to solve are not Physics problems. Generally are change-management and lifecycle-management problems

Reach out to the Experts.



Not Him!

IQC – Institute of Quantum Computing @ University of Waterloo



Quantum Safe security experts like ID Quantique

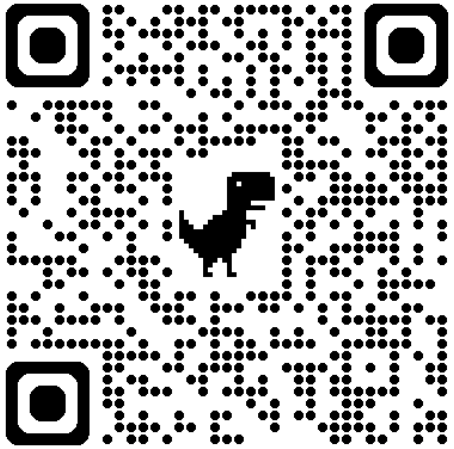
IDQ
an IonQ company

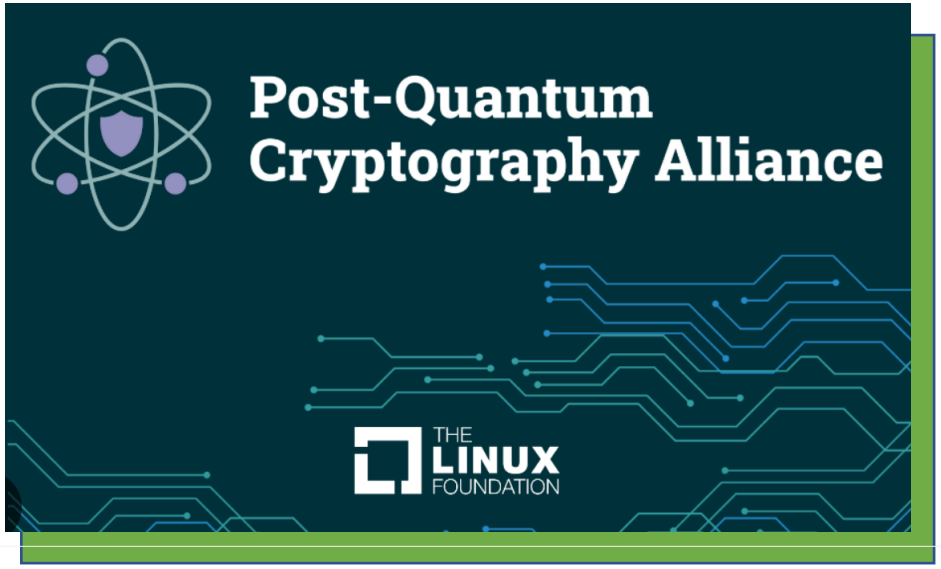
Industry partners like IBM who are building quantum capabilities

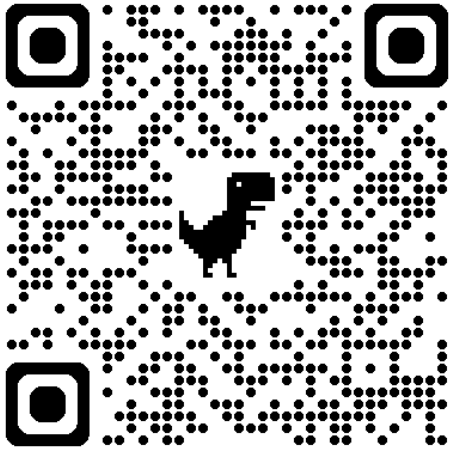


Get Involved in Open Source Projects & Groups.





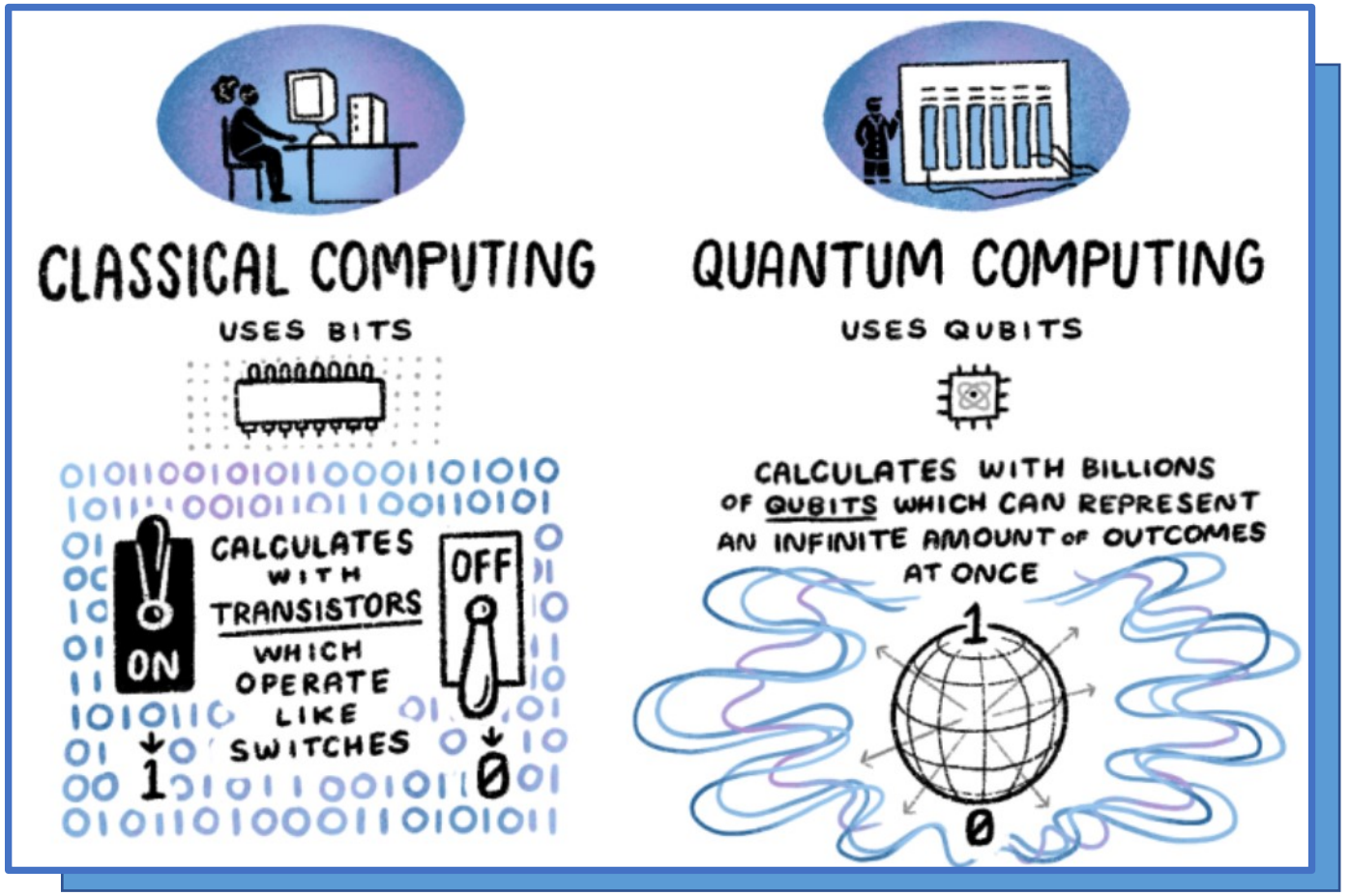




Start an Internal Working Group and do some Info Roadshows.

Find some interested internal groups. Information Security is probably the best group to start with.

Start brainstorming and collaborating then share with your enterprise.



Thank you!

And Please Share Your Feedback.

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